

15.6 Answers

① a) -8

② b) $\frac{1}{3}(e^3 - 1)$

③ a) $\frac{3}{20} \sin 1$

b) $\frac{1}{10}$

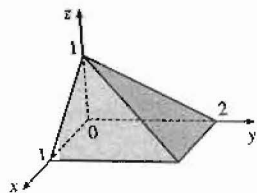
c) $\frac{2}{15}$

d) $\frac{16\pi}{3}$

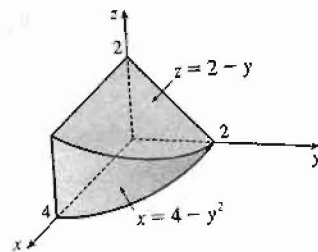
④ a) 144

b) 36π

⑤ a)



b)



⑤

a)

$$\int_{-1}^1 \int_0^{1-x^2} \int_0^y f \, dz \, dy \, dx$$

$$= \int_0^1 \int_{-\sqrt{1-y}}^{\sqrt{1-y}} \int_0^y f \, dz \, dx \, dy$$

$$\int_0^1 \int_0^y \int_{-\sqrt{1-y}}^{\sqrt{1-y}} f \, dx \, dz \, dy$$

$$= \int_0^1 \int_z^{\sqrt{1-y}} \int_{-\sqrt{1-y}}^{\sqrt{1-y}} f \, dx \, dy \, dz$$

$$\int_{-1}^1 \int_0^{1-x^2} \int_z^{1-x^2} f \, dy \, dz \, dx$$

$$= \int_0^1 \int_{-\sqrt{1-z}}^{\sqrt{1-z}} \int_z^{1-x^2} f \, dx \, dz \, dy$$

(6)

$$\int_0^1 \int_0^{y^2} \int_0^{1-y} f \, dz \, dx \, dy$$

$$\int_0^1 \int_0^{1-z} \int_0^{y^2} f \, dx \, dy \, dz$$

$$\int_0^1 \int_0^{1-y} \int_0^{y^2} f \, dx \, dz \, dy$$

$$\int_0^1 \int_0^{1-\sqrt{x}} \int_{\sqrt{x}}^{1-z} f \, dy \, dz \, dx$$

$$\int_0^1 \int_0^{(1-z)^2} \int_{\sqrt{x}}^{1-z} f \, dy \, dx \, dz$$